

STONE CRUSHER PLANT MANAGEMENT SOFTWARE PRODUCT REQUIREMENT DOCUMENT

Version 1.0 | May 2026
Prepared for Client Review

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PRODUCT REQUIREMENT DOCUMENT (PRD)

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1. USER AUTHENTICATION & ACCESS CONTROL

Overview:

Yeh module poore system ka entry point hai. Sirf authorized users hi system access kar sakte hain. Har user ka ek role hoga jo define karega ki woh system mein kya dekh sakta hai aur kya kar sakta hai.

User Roles & Permissions:

Role	Access Level
Admin	Full access - sabhi modules, settings, reports, user management
Accountant	Billing, payments, ledger, expense, reports
Operator	Production entry, machine status, inventory update
Supervisor	Production monitoring, employee attendance, vehicle
Gate Manager	Truck entry/exit, challan generation, weight recording

Features:

- Secure Login: Email/mobile + password based login
- Password Reset: OTP via SMS/email
- Session Management: Auto logout after inactivity
- Role-Based Access: Har module role ke hisaab se lock/unlock hoga
- Activity Logs: Kaun, kab, kya kiya — sab record hoga

2. DASHBOARD

Overview:

Dashboard ek single screen hai jahan plant ka poora status ek nazar mein dikhega. Owner ya Admin subah dashboard kholega aur turant samjhega ki plant ki situation kya hai.

Summary Cards (Real-Time):

- Total Production Today: Kitna material produce hua aaj
- Daily Sales: Aaj kitne ka maal bika
- Pending Payments: Kitna paisa customers se lena baaki hai
- Vehicle Entries Today: Kitne truck aaye/gaye

- Inventory Status: Kaun sa material low hai
- Machine Status: Kaun si machine chal rahi hai, kaun band hai
- Today's Expenses: Aaj kitna kharch hua
- Profit/Loss Overview: Is mahine ka net result

Graphs & Charts:

- Monthly Revenue vs Expense Bar Chart
- Production Trend Line Graph (last 30 days)
- Material-wise Sales Pie Chart
- Vehicle Entry Frequency Chart

Filters:

- Date range filter (Today / This Week / This Month / Custom)

3. INVENTORY MANAGEMENT

Overview:

Plant mein kaafi materials hote hain — raw stone aata hai, crushing hoti hai, aur multiple grades ka finished product nikalta hai. Yeh module poori stock ka hisaab rakhega.

Material Categories:

Category	Type
Raw Stone	Raw Material (Input) ▾
Dust	Finished Product ▾
Sand	Finished Product ▾
10mm Gitti	Finished Product ▾
20mm Gitti	Finished Product ▾
40mm Gitti	Finished Product ▾

Features:

- Stock Entry: Har material ki entry (quantity, unit, date, source)
- Stock In/Out Tracking: Kab material aaya, kab gaya, kitna bacha
- Warehouse/Godown Tracking: Multiple storage locations support
- Low Stock Alert: Minimum level se neeche aane par alert
- Inventory History: Poora audit trail — koi bhi change track hoga
- Edit/Delete with Log: Changes record honge

4. TRUCK & VEHICLE MANAGEMENT

Overview:

Plant par din mein kaafi trucks aate-jaate hain — kuch raw material laate hain, kuch finished product le jaate hain. Gate Manager inka hisaab is module se rakhega.

Entry Data Captured:

Field	Description
Vehicle Number	Truck ka registration number ▾
Driver Name	Driver ka naam ▾
Material Type	Kaun sa material load hai ▾
Weight (Entry)	Truck ka weight aate time ▾
Weight (Exit)	Truck ka weight jaate time ▾
Net Weight	Exit - Entry = actual material weight ▾
Entry Time	Timestamp ▾
Exit Time	Timestamp ▾
Type	Material aa raha hai ya ja raha hai ▾

Features:

- Challan Generation: Har truck trip ka digital challan print/share hoga
- Diesel Tracking: Plant ke trucks ka diesel record
- Vehicle History: Ek truck ki poori trip history
- Weighbridge Integration (Optional): Automatic weight capture

5. BILLING & INVOICE

Overview:

Jab bhi material bechega plant, ek proper GST invoice banana padega. Yeh module billing ka poora kaam karega — invoice banana, payment track karna, aur WhatsApp par share karna.

Invoice Fields:

Field	Description
Invoice No.	Auto-generated, unique (e.g. SCM/2025-26/0001)

Date	Invoice date
Customer Name	From customer master
Address & GSTIN	Auto-filled from customer record
Material	Which product
Quantity	In tons/CFT/units
Rate	Per unit rate
GST %	5% / 12% / 18%
Total Amount	Auto-calculated

Payment Modes:

- Cash: Immediate payment record
- UPI: Reference number capture
- Bank Transfer: NEFT/RTGS — transaction ID record
- Credit: Payment baad mein — outstanding mein jayega

Payment Status: Paid / Partially Paid / Unpaid + Due date

6. CUSTOMER MANAGEMENT

Overview:

Har customer ka ek dedicated record hoga. Koi bhi customer ka naam search karo — pata chalega kitna liya, kitna diya, kitna baaki hai.

Customer Master Data:

- Name, Address, Mobile, GSTIN
- Credit Limit (kitna udhaar de sakte hain)
- Customer Type (Regular / Contractor / One-time)

Customer Ledger:

- Har invoice aur payment ka record
- Running outstanding balance
- Date-wise transaction history

Reports:

- Due Payment Report: Sabhi customers ka outstanding ek jagah
- Customer Purchase Report: Ek customer ne kitna liya

7. EMPLOYEE MANAGEMENT

Overview:

Plant mein operators, loaders, security guards, accountants sab hote hain. Unka attendance, salary, aur shifts is module se manage honge.

Employee Master Data:

Field	Details
Employee ID	Auto-generated
Name	Full name
Contact	Mobile number
Department	Production / Security / Accounts / etc.
Role	Operator / Supervisor / Labour / etc.
Joining Date	Start date
Salary	Monthly / Daily wage

Features:

- Attendance: Daily mark (Present/Absent/Half Day/Leave) + monthly summary
- Salary Management: Auto-calculation based on attendance + salary slip
- Shift Management: Morning/Evening/Night shift assignment

8. PRODUCTION MONITORING

Overview:

Plant mein crusher machine chalti hai aur raw stone se different sizes ka material nikalta hai. Yeh module track karega ki har din kitna production hua, machine kab bandi rahi, aur kyon.

Daily Production Entry Fields:

Field	Details
Date	Production date
Machine Name	Which crusher/unit
Material Type	Which output (dust/10mm/20mm/40mm)
Production Qty	In tons
Working Hours	Machine kitne ghante chali

Downtime	Kitna time band rahi
Downtime Reason	Breakdown/Maintenance/Power Cut/etc.

Features:

- Machine Runtime Tracking: Daily machine hours log
- Downtime Analysis: Kitna production loss hua
- Maintenance Alerts: Jab machine ki service due ho
- Production vs Target: Target vs actual comparison

9. EXPENSE MANAGEMENT

Overview:

Plant ka kharch bohot tarah ka hota hai — diesel, bijli, mazdoori, machine repair. Sabka proper hisaab is module mein hoga.

Expense Categories:

Category	Example
Diesel	Machinery fuel, generator
Electricity	Monthly bill
Labour	Daily wage workers
Maintenance	Machine repair, spare parts
Transportation	Vehicle repair, tyre, oil
Miscellaneous	Office expenses, tea, stationary

Features:

- Expense Entry: Date, category, amount, description, paid by
- Receipt Upload: Photo/PDF attach kar sako bill ki
- Monthly Summary: Category-wise total
- Expense vs Revenue: Profitability check

10. REPORTS & ANALYTICS

Overview:

Month end par ya jab bhi zaroorat ho — ek click mein poori report nikal sako. Sabhi reports PDF aur Excel mein export hoti hain.

Available Reports:

Report	What It Shows
Sales Report	Date/customer/material-wise sales
Inventory Report	Current stock levels, stock movement
Vehicle Report	Truck trips, weight records, challan list
Production Report	Machine-wise, date-wise production summary
Expense Report	Category-wise, month-wise expense
Profit/Loss	Revenue - Expenses = Net Profit
Employee Report	Attendance summary, salary paid
Due Payment	All pending customer payments

Analytics (Visual):

- Monthly Revenue Trend
- Top Customers by Purchase
- Material-wise Sales Analysis
- Expense Breakdown Chart
- Production Efficiency Chart

11. NOTIFICATIONS & ALERTS

Alert Types:

Alert	Trigger
Low Stock Alert	Inventory minimum level se neeche aaye
Payment Due Alert	Customer payment due date aane par
Maintenance Reminder	Machine service schedule ke hisaab se
Invoice Notification	Invoice generate hone par customer ko auto-send

Channels:

- WhatsApp: Invoice sharing, payment reminders
- SMS: Alerts for admin/supervisor
- Email: Reports, invoice copies

12. SECURITY

- Encrypted passwords (no plain text storage)
- Role-based module permissions
- Complete activity log (who did what, when)
- Daily automated database backup
- Secure HTTPS APIs only

13. SYSTEM & DEPLOYMENT

Tech Stack (Recommended):

- Frontend: React.js (responsive, mobile-friendly)
- Backend: Node.js / Laravel
- Database: MySQL / PostgreSQL
- Hosting: VPS or Cloud (AWS / DigitalOcean)
- SSL: Mandatory

Language Support:

- English (default)
- Hindi (toggle option)

Deployment Options:

- Cloud Hosting
- VPS Hosting
- Domain & SSL Setup

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